

GLOWSERVE

Product Requirements Document

At-Home Beauty, Grooming and Wellness Marketplace

CUSTOMER EXPERIENCE

PROFESSIONAL OPERATIONS

ADMIN & MARKETPLACE CONTROL

Technology baseline: PostgreSQL | Express.js | React | Node.js | TypeScript

Document version 1.0 | 30 July 2026

Prepared for portfolio development and implementation planning

Document intent

This PRD defines the product scope, business rules, user journeys, functional and non-functional requirements, data model, API surface, security controls, and delivery plan for an original at-home service marketplace. It does not reproduce any competitor's proprietary interface, copy, data, or internal workflow.

Document Control

Field	Value
Product name	GlowServe
Document type	Product Requirements Document (PRD)
Version	1.0
Status	Implementation baseline
Primary stack	PostgreSQL, Express.js, React, Node.js, TypeScript
Primary applications	Customer web app, Professional app, Admin/Operations app
Audience	Product owner, UI/UX designer, frontend developer, backend developer, QA engineer, DevOps engineer
Delivery strategy	Modular monolith first; service extraction only when operationally justified

Approval and Revision Log

Version	Date	Owner	Summary
1.0	30 Jul 2026	Product / Engineering	Initial complete implementation baseline

How to Use This PRD

- **Requirement IDs:** Use IDs such as AUTH-FR-001 and BOOK-FR-004 in issues, pull requests, test cases, and release notes.
- **Priority:** P0 is mandatory for MVP; P1 is important for marketplace completeness; P2 is an enhancement.
- **Acceptance criteria:** A requirement is complete only when the stated behavior is demonstrated through automated or documented testing.
- **Source of truth:** The server remains authoritative for price, availability, booking status, payment state, permissions, and refunds.

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1. Executive Summary

GlowServe is a multi-role, location-aware marketplace for booking verified beauty, grooming, spa and wellness professionals at a customer's home. The platform coordinates service discovery, pricing, scheduling, professional assignment, payment, OTP-controlled service execution, ratings, customer support, professional earnings, and operational administration.

Core product promise

A customer can find a service available at their address, reserve a valid time slot, complete a transparent checkout, receive a verified professional, track the service lifecycle, and obtain support or a refund through a controlled auditable workflow.

1.1 Product Surfaces

Surface	Primary users	Purpose
Customer application	Customers	Discover, book, pay, track, reschedule, cancel, review and obtain support.
Professional application	Service professionals	Complete KYC, manage availability, accept work, execute services and track earnings.
Admin and operations application	Operations, support, finance, catalog and administrators	Operate the marketplace, resolve exceptions, manage catalog and monitor performance.
Backend platform	All applications and integrations	Enforce business rules, transactions, authorization, event processing and data integrity.

1.2 Product Principles

- Transparent pricing before payment, including taxes, fees, discounts and refund conditions.
- Server-authoritative booking and payment state; client confirmation alone is never sufficient.
- Professional trust through KYC, skill mapping, OTP checkpoints and auditable status history.
- Operational recoverability: unassigned, delayed, cancelled, disputed and failed-payment bookings must remain actionable.
- Permission enforcement on the backend for every protected operation.
- Mobile-first customer and professional experiences; desktop-first administration.

2. Product Vision and Objectives

2.1 Vision

Create a credible, production-oriented service marketplace that can be demonstrated as a full-stack portfolio product and can evolve into a commercially operable platform without replacing its core domain model.

2.2 Product Objectives

- **Reliable booking:** Prevent invalid and overlapping reservations through slot holds, database transactions and conflict checks.
- **High assignment success:** Match confirmed bookings to eligible professionals using skills, location, availability and workload.
- **Safe execution:** Use start and completion OTPs, professional identification, evidence uploads and support escalation.

- **Financial correctness:** Reconcile payment gateway webhooks, calculate refunds, commissions, earnings and settlements deterministically.
- **Operational control:** Provide real-time queues for unassigned, delayed, disputed and financially inconsistent bookings.
- **Portfolio depth:** Demonstrate authentication, RBAC, PostgreSQL design, Socket.IO, queues, testing, Docker and CI/CD.

2.3 Success Metrics

Metric	Definition	Initial target
Booking conversion	Confirmed bookings / checkout sessions	>= 35% in controlled demo testing
Payment confirmation rate	Webhook-verified successful payments / payment attempts	>= 95% excluding simulated failures
Assignment success	Bookings assigned before configured timeout / confirmed bookings	>= 90% in seeded demo environment
Professional acceptance	Accepted requests / delivered assignment requests	>= 70% in seeded demo environment
On-time arrival	Arrivals before scheduled time + tolerance / assigned bookings	>= 85%
Service completion	Completed bookings / started bookings	>= 97%
API reliability	Successful API responses excluding valid 4xx / total requests	>= 99.5% in staging test window
Critical workflow coverage	Automated tests for P0 workflow branches	100% of P0 acceptance paths

3. Users, Personas and Permissions

Persona	Primary needs	Key risks
Customer	Fast service discovery, predictable pricing, trustworthy professional, simple support.	No available slots, hidden charges, late arrival, poor service, refund uncertainty.
Service Professional	Qualified leads, clear job information, safe customer details, reliable earnings.	Unfair assignment, cancellation loss, unsafe location, settlement delay.
Operations Manager	Live visibility, assignment control, exception recovery, customer/professional communication.	Unassigned jobs, delays, incomplete timelines, inconsistent status changes.
Support Executive	Complete booking context, SLA, communication history and resolution tools.	Missing evidence, unauthorized refunds, unresolved escalations.
Finance Manager	Payment reconciliation, refunds, commission and settlement control.	Duplicate payments, incorrect payouts, webhook failures, unresolved chargebacks.
Catalog Manager	Accurate service, pricing, add-on and city availability configuration.	Invalid pricing, inactive service sold, inconsistent service duration.
Super Admin	Secure user, role, configuration and audit management.	Privilege escalation, destructive changes, weak auditability.

3.1 Role Model

Role	Core permissions
Customer	Manage own profile, addresses, cart, bookings, payments, reviews and support tickets.
Professional	Manage own onboarding, availability, assignment responses, service execution and earnings.
Operations Manager	Read and update bookings, assign professionals, manage operational exceptions and contact parties.

Role	Core permissions
Support Executive	Read assigned support context, message users and apply permitted resolutions.
Finance Manager	Read payments, reconcile transactions, approve refunds and process settlements.
Catalog Manager	Create and update catalog, pricing, zones, offers and availability settings.
Super Admin	Manage roles, permissions, platform configuration and high-risk administrative operations.

4. Scope and Release Boundaries

4.1 P0 - Resume-Ready MVP

- Customer, professional and admin authentication with JWT access tokens and rotating refresh sessions.
- Location and address management with serviceability checks.
- Service categories, service details, variants, add-ons and city/zone pricing.
- Cart, coupon validation, server-calculated totals and checkout.
- Availability calculation, temporary slot hold and conflict-safe booking confirmation.
- Online payment order creation, webhook verification and payment reconciliation.
- Professional onboarding, skill mapping, service zones and weekly availability.
- Automatic or manual assignment with request timeout and retry.
- Real-time booking status updates and notification center.
- Start OTP, completion OTP, checklist and evidence upload.
- Customer booking history, cancellation, rating and basic support tickets.
- Admin live operations, booking management, professional management and catalog management.
- Audit logging, Docker, API documentation and automated tests for critical flows.

4.2 P1 - Marketplace Completeness

- Rescheduling with atomic transfer from old slot to new slot.
- Partial and full refunds, refund status timeline and retry operations.
- Professional wallet, commission calculation, settlement periods and payout workflow.
- Membership plans, coupons with usage constraints and customer credits.
- Advanced support SLAs, escalation and resolution types.
- Professional performance scoring, incentives, penalties and suspension workflow.
- Operational reporting, exports and finance reconciliation dashboards.

4.3 P2 - Future Enhancements

- Native mobile applications, live GPS tracking and route optimization.
- Dynamic pricing based on demand, capacity and travel distance.
- Recommendation engine and personalized bundles.
- Automated KYC provider integration and background verification.
- Advanced fraud detection, risk scoring and chargeback management.
- Multi-language content and multi-country taxation.

4.4 Explicitly Out of Scope for MVP

- Copying a competitor's design, copy, private APIs, pricing or operational data.
- Medical diagnosis, clinical services or regulated healthcare workflows.
- Cash collection reconciliation across multiple physical branches.
- Microservices decomposition before validated operational need.
- Fully automated financial settlement without administrative controls.

5. Assumptions and Constraints

ID	Assumption or constraint
A-01	The initial launch supports selected Indian cities and INR pricing.
A-02	A booking contains one address and one scheduled service window, but may contain multiple compatible service items.
A-03	A professional may only receive bookings for approved skills and active service zones.
A-04	The backend is authoritative for availability, price, permission, booking state, payment state and refund eligibility.
A-05	Payment success is confirmed only after verified gateway webhook processing or explicit reconciliation.
A-06	Sensitive documents are stored in object storage using private access and time-limited signed URLs.
A-07	The platform starts as a modular monolith with PostgreSQL as the system of record.
A-08	Redis is available for cache, distributed locks, job queues, acceptance timers and slot-hold expiration.
A-09	Customer and professional applications must work effectively on 360 px mobile screens.
A-10	All high-risk administrative actions must produce immutable audit records.

6. End-to-End User Journeys

6.1 Customer Booking Journey

1. Authenticate or continue browsing as a guest where supported.
2. Select or detect a serviceable location.
3. Browse categories, compare services, inspect inclusions and add compatible items to the cart.
4. Select an address, date and available time slot.
5. Create a temporary slot hold with an expiration timer.
6. Apply eligible coupon or membership benefits.
7. Review a server-calculated price breakdown and cancellation policy.
8. Complete payment or select an enabled pay-after-service method.
9. Receive a confirmed booking only after server-side booking and payment validation.
10. Receive professional assignment and track status in real time.
11. Share start OTP after professional arrival and completion OTP after satisfactory completion.
12. Receive invoice, submit rating, rebook or contact support.

6.2 Professional Journey

13. Register and verify contact information.
14. Complete profile, service skills, service zones, availability, KYC and bank details.
15. Receive approval or actionable requests for additional information.
16. Go online and receive eligible booking requests with acceptance countdown.
17. Accept a request, navigate to the customer and mark en route and arrived states.
18. Verify start OTP, perform checklist-controlled service, record approved add-ons and evidence.
19. Verify completion OTP and submit completion record.
20. View pending earnings, available balance and payout history.

6.3 Operations Exception Journey

21. Monitor unassigned, expiring, delayed and disputed bookings in a real-time queue.
22. Inspect assignment attempts, professional eligibility, payment state and booking timeline.

23. Manually assign, reassign, reschedule, contact parties or escalate to support.
24. Record internal notes and reason codes for every intervention.
25. Close the exception only after the booking reaches a valid terminal or recovered state.

7. Functional Requirements

Priority definitions

P0: required for MVP and portfolio demonstration. P1: required for marketplace completeness. P2: optional enhancement after the core product is stable.

7.1 Authentication and Session Management

Requirement ID	Priority	Requirement	Acceptance criteria
AUTH-FR-001	P0	The system shall support customer and professional registration using verified email or mobile number.	Account remains inactive until the configured verification step succeeds.
AUTH-FR-002	P0	The system shall issue a short-lived JWT access token and a rotating refresh token after successful authentication.	Expired access tokens can be refreshed once per valid refresh-token chain; reused revoked tokens terminate the session family.
AUTH-FR-003	P0	Refresh tokens shall be stored in secure, HTTP-only cookies for browser clients.	JavaScript cannot read the refresh token; production cookies use Secure and an appropriate SameSite policy.
AUTH-FR-004	P0	Users shall be able to log out from the current session or all active sessions.	Revoked sessions cannot obtain a new access token.
AUTH-FR-005	P0	The system shall provide forgot-password and reset-password workflows with expiring single-use tokens.	A used or expired reset token is rejected and all previous password-reset tokens are invalidated.
AUTH-FR-006	P1	The system shall record device/session metadata and allow the user to revoke selected sessions.	The session list shows last activity, approximate device and revocation state without exposing raw tokens.
AUTH-FR-007	P0	Authentication endpoints shall implement rate limiting and failed-login controls.	Repeated attempts trigger progressive protection without permanently locking legitimate users.

7.2 Customer Profile, Location and Address

Requirement ID	Priority	Requirement	Acceptance criteria
CUST-FR-001	P0	Customers shall manage personal information and notification preferences.	Only the customer or an authorized administrator may update the profile.
CUST-FR-002	P0	Customers shall create, edit, label, select and soft-delete addresses.	Deleted addresses remain available in historical bookings but cannot be selected for new bookings.
CUST-FR-003	P0	The platform shall validate city, zone and PIN-code serviceability before showing bookable inventory.	Non-serviceable locations cannot proceed to slot selection and may join a waitlist.

Requirement ID	Priority	Requirement	Acceptance criteria
CUST-FR-004	P1	The platform shall support current-location lookup and map coordinates when location services are available.	The customer can correct the detected address before saving.

7.3 Service Catalog and Pricing

Requirement ID	Priority	Requirement	Acceptance criteria
CAT-FR-001	P0	Administrators shall manage categories, services, variants, add-ons, images, duration, inclusions, FAQs and active status.	Inactive items are unavailable for new bookings but remain visible in historical records.
CAT-FR-002	P0	The platform shall support city- and zone-specific pricing, tax, platform fees and professional earnings.	Checkout uses the pricing version applicable when the slot hold is created.
CAT-FR-003	P0	A service shall declare required professional skills, required products, eligible customer segment and compatibility constraints.	Unavailable or incompatible combinations cannot be added to the same booking.
CAT-FR-004	P1	Catalog managers shall schedule price and availability changes for future activation.	Scheduled versions become active at the configured time without modifying historical booking totals.
CAT-FR-005	P0	Customers shall search, filter and sort services by category, price, duration, rating and availability.	Filters are reflected in API query parameters and return paginated results.

7.4 Cart, Availability and Checkout

Requirement ID	Priority	Requirement	Acceptance criteria
CHECK-FR-001	P0	Customers shall add, update and remove compatible services, variants and add-ons in a persistent server-side cart.	The cart is revalidated when location, address or pricing context changes.
CHECK-FR-002	P0	The server shall calculate subtotal, discounts, taxes, platform fee, travel fee and payable total.	The frontend cannot override any monetary amount.
CHECK-FR-003	P0	The availability service shall calculate slots using professional skills, zone, schedule, existing bookings, service duration and travel buffer.	Returned slots include capacity or availability state and cannot include known conflicts.
CHECK-FR-004	P0	Selecting a slot shall create an expiring slot hold associated with the customer and cart.	An expired hold cannot be converted into a booking and its capacity is released.
CHECK-FR-005	P0	The system shall detect price or availability changes before payment confirmation.	The customer must explicitly accept an updated total or select a different slot.
CHECK-FR-006	P1	The platform shall support automatic assignment and optional customer-selected professional modes.	Only professionals eligible for every required service are selectable.

7.5 Booking Lifecycle

Requirement ID	Priority	Requirement	Acceptance criteria
BOOK-FR-001	P0	The system shall create a booking and booking items within a PostgreSQL transaction.	Partial bookings are rolled back if any required insert, constraint or payment precondition fails.
BOOK-FR-002	P0	Every booking status transition shall be validated against an explicit state-transition map.	Invalid direct transitions return a conflict response and create no side effects.
BOOK-FR-003	P0	Every transition shall append an immutable status-history record with actor, timestamp, source and reason.	The booking timeline can reconstruct the complete lifecycle.
BOOK-FR-004	P0	Customers shall view upcoming, active, completed and cancelled bookings.	Each booking exposes only permitted actions based on state and policy.
BOOK-FR-005	P0	Eligible bookings shall support cancellation with fee and refund calculation.	The customer sees the fee and estimated refund before confirmation.
BOOK-FR-006	P1	Eligible bookings shall support rescheduling using a held new slot and atomic release of the old slot.	The original reservation remains valid until the new slot is successfully confirmed.
BOOK-FR-007	P0	The system shall detect customer no-show and professional no-show outcomes.	The booking stores the responsible party, evidence and financial resolution.

7.6 Professional Onboarding and Availability

Requirement ID	Priority	Requirement	Acceptance criteria
PRO-FR-001	P0	Professionals shall complete a resumable multi-step onboarding application.	Draft progress is preserved and validation is applied per step and on final submission.
PRO-FR-002	P0	Professionals shall upload required KYC, certification and bank documents.	Files are private, virus-scanned where available and accessible only through authorized signed URLs.
PRO-FR-003	P0	Authorized administrators shall approve, reject or request additional information for each verification area.	Every decision records reviewer, reason, timestamp and affected document.
PRO-FR-004	P0	Professionals shall define recurring availability, breaks, time off, blackout dates and service zones.	Assignment excludes unavailable intervals and suspended zones.
PRO-FR-005	P0	Only approved, active and sufficiently skilled professionals may receive assignment requests.	The eligibility query excludes incomplete KYC, suspended accounts and missing skills.
PRO-FR-006	P1	Professionals shall control temporary online/offline availability.	Going offline prevents new requests but does not cancel accepted bookings.

7.7 Assignment Engine

Requirement ID	Priority	Requirement	Acceptance criteria
ASSIGN-FR-001	P0	The system shall find eligible professionals using required skills, zone, availability and conflict checks.	The candidate set contains no ineligible professional.

Requirement ID	Priority	Requirement	Acceptance criteria
ASSIGN-FR-002	P0	Eligible professionals shall be ranked using configurable distance, rating, acceptance, completion and workload factors.	The scoring components and final score are logged for operational review.
ASSIGN-FR-003	P0	The system shall send a time-limited assignment request to the highest-ranked candidate.	Expired requests cannot be accepted and trigger the next assignment attempt.
ASSIGN-FR-004	P0	Acceptance shall atomically assign the professional and prevent a second professional from accepting the same booking.	Concurrent accepts result in exactly one successful assignment.
ASSIGN-FR-005	P0	Operations users shall manually assign or reassign an eligible professional with a reason.	Manual override validates conflicts and creates an audit entry.
ASSIGN-FR-006	P1	Assignment failures shall enter an operational exception queue.	The queue exposes elapsed time, attempts, candidate count and recommended action.

7.8 Service Execution

Requirement ID	Priority	Requirement	Acceptance criteria
EXEC-FR-001	P0	Professionals shall update en-route and arrived states for assigned bookings.	The customer receives real-time notification and the status history records the professional actor.
EXEC-FR-002	P0	The system shall generate an expiring start OTP after arrival.	Service cannot enter IN_PROGRESS until the correct OTP is verified or an authorized override is recorded.
EXEC-FR-003	P0	The professional shall complete required service checklists and may record approved add-ons.	Additional charges require customer confirmation before they affect the final amount.
EXEC-FR-004	P0	The system shall generate an expiring completion OTP.	Booking cannot enter COMPLETED without successful OTP verification or documented authorized override.
EXEC-FR-005	P1	Professionals shall upload permitted before/after evidence and incident reports.	Evidence access follows role and retention rules.
EXEC-FR-006	P0	Completion shall trigger invoice generation, earnings calculation and review eligibility.	Each side effect is idempotent and retry-safe.

7.9 Payments and Refunds

Requirement ID	Priority	Requirement	Acceptance criteria
PAY-FR-001	P0	The backend shall create payment orders using the current server-calculated payable amount.	The gateway order amount and internal amount match exactly.
PAY-FR-002	P0	The backend shall verify payment webhook signatures and store processed gateway event IDs.	Invalid signatures are rejected and duplicate events are idempotently ignored.
PAY-FR-003	P0	A successful frontend payment callback shall not independently confirm payment.	The booking waits for verified webhook or explicit reconciliation.

Requirement ID	Priority	Requirement	Acceptance criteria
PAY-FR-004	P0	Payment and booking confirmation operations shall be idempotent.	Retrying the same request cannot create duplicate bookings or payments.
PAY-FR-005	P1	Authorized users shall create partial or full refunds within policy limits.	Refund amount cannot exceed captured amount minus previous refunds.
PAY-FR-006	P1	Refund state shall be updated from verified gateway events and shown to the customer.	Customer sees initiated, processing, completed or failed status with timestamps.
PAY-FR-007	P1	Finance users shall reconcile unmatched or inconsistent payment events.	Manual resolution requires reason and audit trail.

7.10 Wallet, Commission and Settlements

Requirement ID	Priority	Requirement	Acceptance criteria
WALLET-FR-001	P1	Completion shall create immutable professional earning ledger entries.	Ledger totals reconcile to booking items, commission, incentives, penalties, tax and refund adjustments.
WALLET-FR-002	P1	Earnings shall move from pending to available after the configured dispute period.	Disputed or refunded bookings remain frozen or adjusted.
WALLET-FR-003	P1	Professionals shall request payout from available balance subject to thresholds and verification.	The same balance cannot be paid twice.
WALLET-FR-004	P1	Finance users shall review, approve, process and retry settlements.	Every settlement has a deterministic status timeline and bank reference where available.

7.11 Coupons, Membership and Credits

Requirement ID	Priority	Requirement	Acceptance criteria
PROMO-FR-001	P1	Administrators shall configure coupons by dates, usage limits, user eligibility, city, category, service and minimum value.	A coupon is applied only when every rule succeeds.
PROMO-FR-002	P1	The system shall prevent duplicate redemption beyond per-user and global limits.	Concurrent redemption requests cannot exceed configured usage.
PROMO-FR-003	P1	Customers shall purchase and view membership plans with explicit benefits and renewal state.	Benefits are applied by the server and shown separately in the price breakdown.
PROMO-FR-004	P1	Authorized support users shall issue expiring platform credits within role-based limits.	Issued credits include reason, source ticket and audit record.

7.12 Reviews and Quality

Requirement ID	Priority	Requirement	Acceptance criteria
REV-FR-001	P0	Only customers with completed eligible bookings shall submit one review per booking.	Duplicate reviews are prevented and edits follow configured policy.
REV-FR-002	P0	Reviews shall support overall rating and quality dimensions such as behavior, hygiene and punctuality.	Aggregate professional and service ratings are recalculated asynchronously.

Requirement ID	Priority	Requirement	Acceptance criteria
REV-FR-003	P1	Administrators shall moderate flagged reviews without modifying the customer's original text silently.	Hidden or removed reviews retain moderation reason and audit history.

7.13 Support and Disputes

Requirement ID	Priority	Requirement	Acceptance criteria
SUP-FR-001	P0	Customers and professionals shall create support tickets linked to a booking or account issue.	The ticket captures category, priority, description, evidence and contact preference.
SUP-FR-002	P0	Support queues shall assign ownership and track SLA deadlines.	Overdue tickets are visibly escalated.
SUP-FR-003	P1	Authorized support roles shall apply defined resolutions such as refund, credit, repeat service or professional warning.	Resolution actions validate role limits and create linked financial or operational records.
SUP-FR-004	P0	Support users shall access a consolidated booking, payment, assignment, message and evidence timeline.	The support view contains enough context to decide without querying production databases manually.

7.14 Notifications and Real-Time Updates

Requirement ID	Priority	Requirement	Acceptance criteria
NOTIF-FR-001	P0	The platform shall create in-app notifications for material booking, payment, refund, support and account events.	Notifications support read/unread state and deep links.
NOTIF-FR-002	P0	Socket.IO shall deliver real-time booking and assignment updates to authorized rooms.	A user receives only events for permitted entities.
NOTIF-FR-003	P0	Email/SMS/push delivery shall use background jobs with retry and dead-letter handling.	Transient failures retry without duplicating user-visible notifications.
NOTIF-FR-004	P1	Users shall configure channel preferences for non-mandatory notifications.	Mandatory security and transaction notifications cannot be fully disabled.

7.15 Admin, Operations and Reporting

Requirement ID	Priority	Requirement	Acceptance criteria
ADMIN-FR-001	P0	Operations users shall view live queues for unassigned, delayed, cancelled, failed-payment and disputed bookings.	Queues update in real time and support filters and detail drawers.
ADMIN-FR-002	P0	Admin tables shall support search, filters, sorting, pagination, exports and permission-controlled row actions.	Large datasets are processed server-side.
ADMIN-FR-003	P0	Catalog users shall manage service hierarchy, prices, zones and activation state.	Changes are validated against active future bookings where relevant.
ADMIN-FR-004	P1	Finance dashboards shall expose payment, refund and settlement reconciliation states.	Every mismatch is actionable and exportable.

Requirement ID	Priority	Requirement	Acceptance criteria
ADMIN-FR-005	P1	Reports shall provide booking, revenue, cancellation, professional utilization and customer-retention metrics.	Metrics specify timezone, date range and calculation definition.

7.16 RBAC, Audit and Configuration

Requirement ID	Priority	Requirement	Acceptance criteria
RBAC-FR-001	P0	The platform shall support roles mapped to granular permissions.	Backend authorization checks the required permission and relevant resource scope.
RBAC-FR-002	P0	Administrative users shall not be able to grant permissions they do not possess unless they are Super Admin.	Privilege escalation attempts are rejected and logged.
AUDIT-FR-001	P0	High-risk changes shall create immutable audit events containing actor, action, entity, before/after reference, IP and request ID.	Audit records are searchable and cannot be edited through normal application endpoints.
CONFIG-FR-001	P1	Authorized administrators shall manage operational settings such as slot-hold duration, assignment timeout, travel buffer and cancellation windows.	Configuration changes are versioned and take effect predictably.

8. Business Rules

ID	Rule	Definition
BR-001	Price authority	All totals are calculated by the backend from versioned pricing, active promotions and customer context.
BR-002	Historical integrity	Completed or cancelled bookings retain the exact service description, duration, price, tax and policy snapshot used at purchase.
BR-003	Slot ownership	A slot hold belongs to one authenticated customer/session and cannot be transferred.
BR-004	Conflict protection	A professional cannot hold overlapping assigned or active bookings after travel buffers are applied.
BR-005	Assignment eligibility	Professional must be active, approved, online where required, skilled for every service and serviceable for the address.
BR-006	OTP override	Only specifically authorized roles may override OTP checkpoints, and the override requires a reason and audit event.
BR-007	Cancellation policy	Cancellation fee depends on actor, time to booking, professional travel state, service start state and configured policy version.
BR-008	Refund limit	Cumulative successful and pending refunds cannot exceed captured payment.

ID	Rule	Definition
BR-009	Review eligibility	Only the customer attached to a completed booking may review it.
BR-010	Settlement freeze	Disputed, refunded or chargeback-affected earnings remain unavailable until resolution.
BR-011	Data scope	Admin access is limited by permissions and optional city/region scope.
BR-012	Deletion	Operational and financial records are retained; user-facing deletion is implemented through deactivation, anonymization or soft deletion where legally appropriate.

9. Booking State Machine

Controlled Booking State Machine

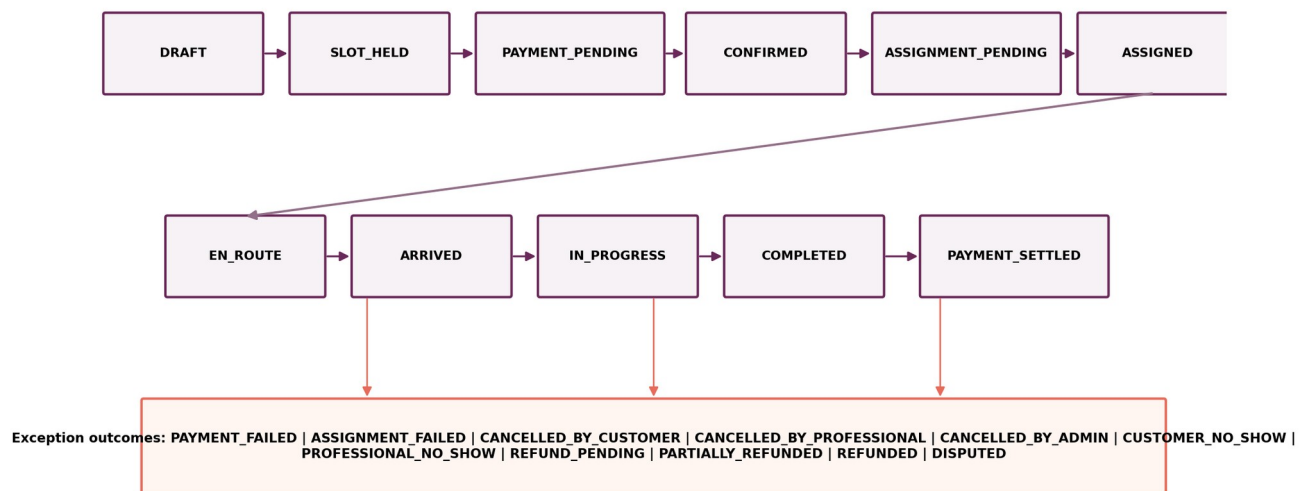


Figure 1. Booking lifecycle and exception outcomes.

9.1 Valid Primary Transitions

From	To	Trigger
DRAFT	SLOT_HELD	Customer selects a valid slot and the server creates a hold.
SLOT_HELD	PAYMENT_PENDING	Checkout creates an online payment order.
SLOT_HELD	CONFIRMED	Enabled pay-after-service method passes confirmation checks.
PAYMENT_PENDING	CONFIRMED	Verified successful payment event and booking transaction complete.
CONFIRMED	ASSIGNMENT_PENDING	Assignment workflow starts.
ASSIGNMENT_PENDING	ASSIGNED	Exactly one eligible professional accepts or is manually assigned.
ASSIGNED	PROFESSIONAL_EN_ROUTE	Professional starts travel.
PROFESSIONAL_EN_ROUTE	PROFESSIONAL_ARRIVED	Professional confirms arrival.
PROFESSIONAL_ARRIVED	IN_PROGRESS	Start OTP or authorized override succeeds.

From	To	Trigger
IN_PROGRESS	COMPLETED	Completion checklist and completion OTP succeed.
COMPLETED	PAYMENT_SETTLED	All payment/earning conditions reach settlement-ready or settled state.

9.2 Transition Enforcement

- Use a single domain service to validate transitions rather than updating status fields directly in controllers.
- Each transition writes booking_status_history in the same transaction as the booking status update.
- Publish domain events through an outbox or retry-safe event mechanism after the transaction commits.
- Use idempotency keys for payment confirmation, completion and administrative actions that may be retried.

10. Data Model

PostgreSQL is the system of record. Monetary values should be stored as integer minor units or exact numeric types with explicit currency. Timestamps should be stored in UTC and converted at presentation boundaries.

10.1 Core Entities

Domain	Tables
Identity	users, user_sessions, roles, permissions, role_permissions, user_roles
Customer	customer_profiles, customer_addresses, customer_credits
Professional	professional_profiles, professional_documents, professional_skills, professional_service_zones, professional_availability, professional_time_off
Catalog	cities, service_zones, categories, services, service_variants, service_addons, service_prices, service_price_versions
Commerce	carts, cart_items, slot_holds, bookings, booking_items, booking_status_history
Assignment	assignment_attempts, booking_assignments, professional_location_snapshots
Execution	booking_otps, service_checklists, booking_evidence, booking_incidents
Payments	payment_orders, payments, payment_events, refunds, invoices
Professional finance	professional_wallets, wallet_transactions, settlements, settlement_items
Growth	coupons, coupon_rules, coupon_redemptions, memberships, customer_memberships
Quality and support	reviews, review_dimensions, support_tickets, support_messages, support_actions
Platform	notifications, notification_preferences, audit_logs, outbox_events, system_config_versions

10.2 Key Relationships

- A user may have one customer profile, one professional profile, and multiple administrative role assignments.
- A service may have multiple variants, add-ons, required skills and versioned prices by city or zone.
- A booking owns immutable booking items and references a pricing/policy snapshot.
- A booking may have multiple assignment attempts but at most one active accepted assignment.
- A payment order may receive multiple gateway events, but processed event IDs are unique.
- A refund belongs to a captured payment and may be partial; cumulative refunds are constrained.
- Wallet transactions are append-only ledger entries and settlement items reference the ledger entries they pay.

10.3 Required Constraints and Indexes

Object	Requirement
users	Unique normalized email and/or phone; status index.
user_sessions	Unique token-family identifier; indexes by user and expiration.
service_prices	Unique active price context per service/variant/add-on, location and effective window.
slot_holds	Index by expiration and customer; state prevents conversion after expiry.
booking_assignments	Partial unique index allowing one active accepted assignment per booking.
professional bookings	Range conflict strategy using time ranges, locking, or exclusion constraints where practical.
payment_events	Unique gateway + gateway_event_id for webhook idempotency.
coupon_redemptions	Unique constraints and transaction-safe counters for per-user and global limits.
reviews	Unique booking_id + customer_id.
audit_logs	Indexes by actor, action, entity, entity_id and created_at; no update/delete through application role.

11. API Requirements

11.1 API Conventions

- Base path: /api/v1.
- JSON request and response bodies; multipart only for file upload endpoints.
- Standard error envelope containing code, message, field errors, request ID and safe metadata.
- Cursor pagination for activity feeds and large timelines; page pagination acceptable for admin tables where count is required.
- Idempotency-Key header for booking creation, payment confirmation, refund creation and payout requests.
- OpenAPI documentation generated and validated in CI.

Method	Endpoint	Purpose
POST	/auth/register	Create customer or professional identity.
POST	/auth/login	Authenticate and issue session.
POST	/auth/refresh	Rotate refresh token and issue new access token.
POST	/auth/logout	Revoke current session.
GET	/catalog/categories	List active categories for location.
GET	/catalog/services	Search and filter active services.
GET	/services/{id}	Get service detail and eligible variants/add-ons.
GET	/availability	Calculate bookable slots for cart and address.
POST	/slot-holds	Create expiring slot hold.
GET	/cart	Get current cart.
POST	/cart/items	Add validated item.
PATCH	/cart/items/{id}	Update quantity, variant or add-ons.
POST	/checkout/quote	Return authoritative checkout quote.
POST	/payments/orders	Create gateway order.
POST	/payments/webhooks/{gateway}	Receive and verify gateway event.
POST	/bookings	Create/confirm booking idempotently.
GET	/bookings	List current user bookings.
GET	/bookings/{id}	Get authorized booking detail and timeline.
POST	/bookings/{id}/cancel	Cancel according to policy.
POST	/bookings/{id}/reschedule	Hold and confirm new slot.
GET	/professional/requests	List assignment requests.
POST	/professional/requests/{id}/accept	Atomically accept assignment.

Method	Endpoint	Purpose
POST	/professional/requests/{id}/reject	Reject with reason.
POST	/professional/bookings/{id}/status	Execute controlled status transition.
POST	/professional/bookings/{id}/verify-start	Verify start OTP.
POST	/professional/bookings/{id}/verify-completion	Verify completion OTP.
POST	/reviews	Create eligible booking review.
POST	/support/tickets	Create support ticket.
GET	/admin/operations/queues	Get operational exception queues.
POST	/admin/bookings/{id}/assign	Manual assignment.
GET	/admin/reports/{report}	Get permission-scoped report.

11.2 Standard Error Codes

Code	Meaning
AUTH_INVALID_CREDENTIALS	Credentials are invalid without revealing which field failed.
AUTH_SESSION_REVOKED	Refresh session or token family is no longer valid.
PERMISSION_DENIED	Authenticated user lacks required permission or resource scope.
LOCATION_NOT_SERVICEABLE	Selected address cannot currently be served.
SLOT_UNAVAILABLE	Slot expired, reached capacity or conflicts with another confirmed action.
PRICE_CHANGED	Authoritative quote differs from the customer-accepted quote.
INVALID_BOOKING_TRANSITION	Requested state transition is not allowed.
PAYMENT_PENDING_VERIFICATION	Gateway callback received but verified payment event is not yet complete.
PAYMENT_EVENT_INVALID	Webhook signature or payload validation failed.
REFUND_LIMIT_EXCEEDED	Requested refund exceeds remaining captured amount.
IDEMPOTENCY_CONFLICT	The same idempotency key was reused with different request content.

12. System Architecture

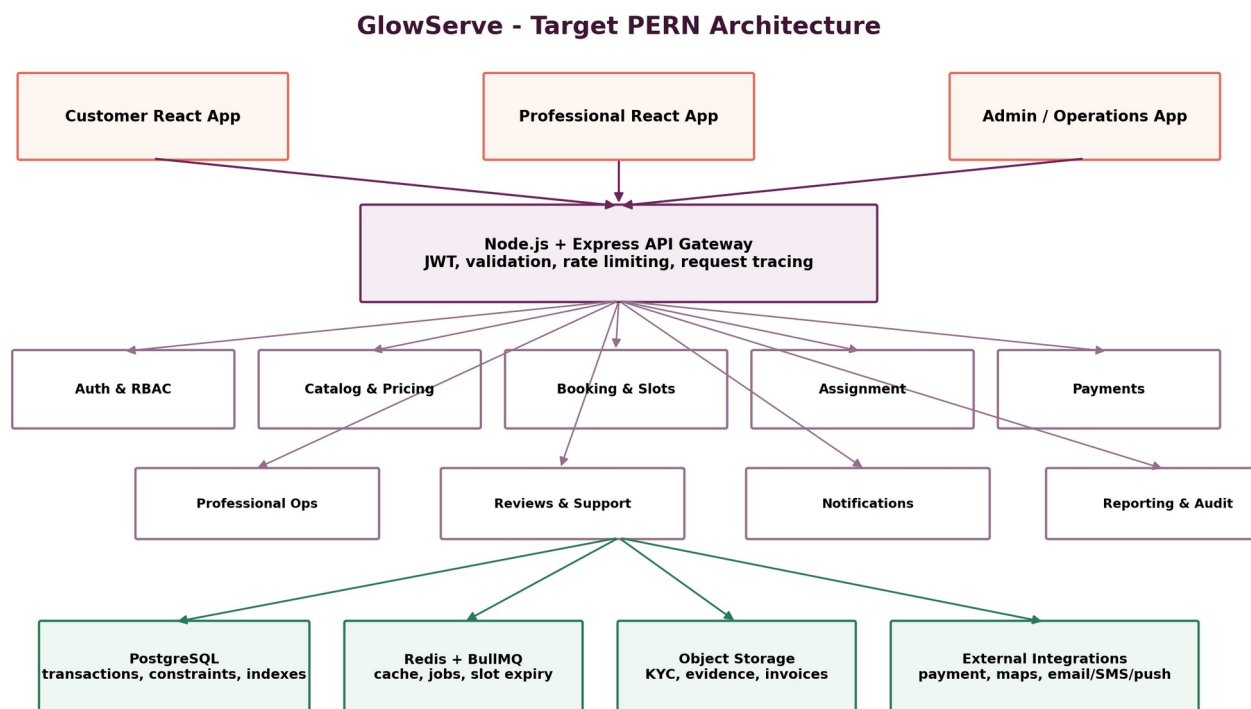


Figure 2. Target modular-monolith architecture and external dependencies.

12.1 Architectural Decisions

Decision	Rationale
Modular monolith	Provides clear domain boundaries without premature distributed-system complexity.
PostgreSQL system of record	Supports transactions, constraints, relational reporting and conflict-safe booking logic.
Redis + BullMQ	Supports expiring holds, assignment timers, notification retries, scheduled reminders and cache.
Socket.IO	Provides real-time booking, assignment and support updates with authenticated rooms.
Object storage	Separates binary KYC/evidence/invoice files from application database storage.
Outbox/retry-safe events	Prevents lost side effects when a database transaction succeeds but downstream delivery fails.
Versioned business configuration	Preserves reproducibility for pricing, cancellation and commission calculations.

12.2 Suggested Backend Module Boundaries

- auth, users, sessions and RBAC
- customers, addresses and location serviceability
- catalog, pricing and promotions
- carts, availability, slot holds and checkout
- bookings, status transitions and rescheduling
- professionals, KYC, skills and availability
- assignment and operational exception handling

- payments, refunds, invoices and reconciliation
- wallet, commission and settlements
- reviews, support, notifications and audit

13. Non-Functional Requirements

ID	Area	Priority	Requirement
NFR-001	Availability	P0	Target 99.5% monthly availability for staging/production demonstration excluding planned maintenance.
NFR-002	API latency	P0	95th percentile under 500 ms for standard reads and under 900 ms for standard writes, excluding third-party latency.
NFR-003	Search latency	P1	Catalog and admin search responses under 800 ms p95 for expected indexed dataset.
NFR-004	Scalability	P0	Stateless API instances shall support horizontal scaling; jobs and sockets shall use shared Redis coordination.
NFR-005	Consistency	P0	Booking, assignment acceptance, payment confirmation, refunds and settlements use transaction or idempotency controls.
NFR-006	Durability	P0	Automated PostgreSQL backups and tested restore procedure for production-like deployment.
NFR-007	Security	P0	OWASP-aligned controls, dependency scanning, secret management and secure headers.
NFR-008	Accessibility	P0	Customer and professional critical flows conform to WCAG 2.2 AA principles.
NFR-009	Compatibility	P0	Latest two stable versions of Chrome, Edge, Firefox and Safari; responsive support from 360 px.
NFR-010	Observability	P0	Structured logs, request IDs, error tracking, metrics and alertable critical workflow failures.
NFR-011	Maintainability	P0	Strict TypeScript, module boundaries, linting, formatting and CI-enforced test checks.
NFR-012	Privacy	P0	Least-privilege access, retention rules and secure private file access.
NFR-013	Localization	P1	Currency, time zone and user-visible date formatting shall be centralized and localization-ready.
NFR-014	Degraded operation	P1	Users receive explicit pending/retry states during gateway, socket or notification outages.

14. Security and Privacy

14.1 Security Controls

Control area	Requirement
Authentication	Strong password hashing, expiring tokens, refresh rotation, session revocation, optional MFA.
Authorization	Backend RBAC plus resource ownership and optional city/region scope.
Input security	Schema validation, output encoding, parameterized SQL/ORM queries and upload validation.
Web security	Secure headers, CORS allowlist, CSRF strategy for cookie endpoints and XSS-resistant rendering.
Rate limiting	Per-IP and per-account limits on authentication, OTP, checkout, support and administrative endpoints.
Secrets	Environment or secret-manager storage; no secrets in client bundles or source control.
Files	Private buckets, signed URLs, content-type validation, size limits and malware scanning where available.
Payments	No raw card storage; verify gateway signatures and restrict sensitive logs.
Audit	Append-only high-risk action records with request ID, actor and reason.
Data exposure	Professional receives customer contact/address only after assignment and only as required for fulfillment.

14.2 Privacy and Retention

- Define retention periods for KYC documents, OTP records, precise location snapshots, support evidence and logs.
- Provide account deactivation and a documented deletion/anonymization workflow consistent with financial retention obligations.
- Redact sensitive fields from application logs and error tracking.
- Use purpose-limited access for support, finance, KYC and operations roles.
- Expose privacy notices and consent for location, camera, file upload and background verification where applicable.

15. Analytics and Observability

15.1 Product Events

Event	Required properties
location_selected	user_id/session_id, city, zone, serviceable
service_viewed	service_id, category_id, location, source
cart_item_added	service_id, variant_id, add_on_ids, amount
slot_hold_created	hold_id, slot, duration, cart_value
checkout_started	cart_id, quote_id, total
payment_order_created	payment_order_id, gateway, amount
booking_confirmed	booking_id, amount, service_count, assignment_mode
assignment_request_sent	booking_id, professional_id, rank, score
assignment_accepted	booking_id, professional_id, response_seconds
service_started	booking_id, professional_id, delay_minutes
service_completed	booking_id, actual_duration, final_amount
booking_cancelled	booking_id, actor, reason, fee, refund
review_submitted	booking_id, service_rating, professional_rating
support_ticket_created	ticket_id, booking_id, category, priority

15.2 Operational Monitoring

- API request count, latency, status code and error code by endpoint.
- Database connection usage, slow queries, lock waits and transaction rollbacks.
- Queue depth, job age, retries and dead-letter count by job type.
- Socket connection count and failed room authorization.
- Payment webhook failures, duplicates, unmatched events and reconciliation backlog.
- Unassigned booking age, assignment attempt count and acceptance timeout rate.
- Refund and settlement failures requiring manual intervention.

16. UX and Accessibility

16.1 Responsive Strategy

Application	Primary breakpoint strategy
Customer	Mobile-first at 360/390 px; tablet at 768 px; desktop enhancements at 1024/1440 px.
Professional	Mobile-first; critical actions usable with one hand and limited connectivity.
Admin	Desktop-first with responsive filters, horizontal table handling and detail drawers.

16.2 Required UX States

- Loading, skeleton, empty, permission denied, validation error, API error and offline/degraded states.
- Location not serviceable, no slots, slot expired, price changed and no eligible professional states.
- Payment failed, payment pending verification, refund pending and reconciliation-required states.
- Assignment delayed, professional unavailable, customer no-show and professional no-show states.
- Explicit confirmation dialogs for cancellation, refund, manual assignment, suspension and destructive catalog changes.

16.3 Accessibility Requirements

- Keyboard-operable navigation, forms, dialogs, menus, tables and time-slot selection.
- Visible focus indicators and semantic headings.
- Labels and accessible descriptions for form controls and status updates.
- Status must not rely on color alone.
- Error summaries plus field-level errors.
- Minimum touch target sizes and reduced-motion support.
- Accessible live regions for payment, assignment and booking status updates.

17. Testing and Acceptance

17.1 Test Strategy

Layer	Scope	Suggested tools
Unit	Domain calculations, validators, transition rules, assignment scoring, cancellation and commission logic.	Vitest or Jest
API integration	Auth, booking transactions, concurrency, permissions, payment webhooks, refunds and persistence.	Supertest + test PostgreSQL
Component	Forms, slot picker, price breakdown, data tables and permission states.	React Testing Library

Layer	Scope	Suggested tools
End-to-end	Customer booking, professional acceptance, OTP completion, cancellation and admin recovery.	Playwright
Load/concurrency	Slot holds, simultaneous assignment acceptance, webhook bursts and report endpoints.	k6 or Artillery
Security	Dependency scanning, authorization checks, input attacks, upload restrictions and secret scanning.	CI scanners + manual checklist

17.2 Critical Acceptance Scenarios

ID	Acceptance scenario
AC-01	Two customers attempt the final available capacity for the same slot; only one confirmation succeeds and no duplicate booking is created.
AC-02	Two professionals accept the same assignment request concurrently; exactly one is assigned.
AC-03	A verified payment webhook is delivered twice; payment and booking are confirmed once.
AC-04	A frontend success callback arrives but webhook verification is delayed; the UI shows pending verification and does not falsely confirm.
AC-05	A slot hold expires before payment; booking creation fails with SLOT_UNAVAILABLE and the hold capacity is released.
AC-06	A customer attempts an invalid booking status action or accesses another customer's booking; the request is denied.
AC-07	A professional without the required skill or zone is excluded from assignment and cannot be manually assigned.
AC-08	Completion is retried after a network timeout; invoice and wallet entries are created once.
AC-09	A partial refund plus a later refund cannot exceed the captured payment amount.
AC-10	An unauthorized support user attempts a refund above their limit; the operation is rejected and audited.
AC-11	A reschedule fails after the new hold is created; the original booking slot remains intact.
AC-12	Every PO requirement has at least one mapped automated or documented test case before release.

17.3 Definition of Done

- Functional behavior and acceptance criteria are complete.
- API contract and validation are documented.
- Permission checks exist on backend and corresponding UI state exists on frontend.
- Loading, empty, error and success states are implemented.
- Automated tests pass in CI and no critical accessibility regression remains.
- Logs, metrics and audit events are present for operationally significant behavior.
- Documentation and demo seed data are updated.

18. Release Plan

Milestone	Indicative duration	Deliverables
Milestone 1 - Foundation	Weeks 1-2	Repository, Docker, CI, PostgreSQL schema, authentication, sessions, RBAC, shared UI and API error standards.

Milestone	Indicative duration	Deliverables
Milestone 2 - Catalog and Customer Core	Weeks 3-4	Location, addresses, catalog, pricing, cart, quote and customer profile.
Milestone 3 - Booking and Payment	Weeks 5-7	Availability, slot holds, booking transaction, payment orders, webhook verification and booking history.
Milestone 4 - Professional Operations	Weeks 8-10	Onboarding, KYC, skills, availability, assignment, status updates and OTP execution.
Milestone 5 - Admin and Real-Time	Weeks 11-12	Operations queues, booking/professional/catalog admin, Socket.IO and notifications.
Milestone 6 - Quality and Deployment	Weeks 13-14	Reviews, support, audit, E2E testing, security hardening, seeded demo and deployment.
Milestone 7 - P1 Expansion	Post-MVP	Rescheduling, advanced refunds, wallet, settlements, promotions, memberships and analytics.

18.1 Deployment Environments

Environment	Purpose	Controls
Local	Developer execution and automated tests.	Docker Compose, seeded data, local mail/payment mocks.
Staging	Integrated QA and portfolio demonstration rehearsal.	Production-like configuration, sandbox gateways, anonymized seed data.
Production/Demo	Public portfolio deployment.	Managed secrets, HTTPS, backups, monitoring, rate limits and restricted admin access.

19. Risks and Mitigations

Risk	Impact	Mitigation
Double booking or capacity race	High	Use transactional confirmation, row/range locks, conflict constraints and concurrency tests.
Payment/booking inconsistency	High	Webhook verification, idempotency, reconciliation queue and clear pending state.
Unauthorized access to customer/KYC data	High	Backend RBAC, resource scope, private files, signed URLs and audit logs.
Assignment failure and customer delay	High	Eligibility diagnostics, multiple attempts, manual queue, notifications and escalation SLA.
Incorrect cancellation/refund amount	High	Versioned policies, pure calculation functions, money precision and approval limits.
Premature architecture complexity	Medium	Start modular monolith; extract services only after measurable operational need.
Admin dashboard becomes unmanageable	Medium	Permission-specific navigation, server-side tables, saved filters and detail drawers.
Third-party outage	Medium	Explicit pending states, retries, circuit-breaker behavior and manual reconciliation.
Portfolio project scope becomes too large	High	Deliver P0 vertical slices first; P1/P2 remain feature flags or roadmap items.

20. Open Decisions

ID	Decision required	Owner
OD-01	Will MVP support pay-after-service, or online payment only?	Product owner
OD-02	Will customers select a professional or use automatic assignment only?	Product owner / Operations
OD-03	What is the initial city, zone and travel-fee model?	Operations / Catalog
OD-04	Which payment gateway will be used for the deployment region?	Engineering / Finance
OD-05	Which KYC documents are mandatory for the demo and how long are they retained?	Operations / Security
OD-06	What cancellation windows and fees apply to each lifecycle state?	Product / Finance
OD-07	Will the professional app expose live customer location/contact before acceptance, after acceptance or only en route?	Security / Operations
OD-08	What settlement frequency and minimum payout threshold will be demonstrated?	Finance
OD-09	Will service bundles allow different professionals or require a single multi-skilled professional?	Product / Assignment
OD-10	Which analytics provider, error tracker and deployment platform will be used?	Engineering

Appendix A. Glossary

Term	Definition
Assignment attempt	A time-limited offer of a confirmed booking to an eligible professional.
Booking item	An immutable service, variant or add-on purchased within a booking.
Captured payment	A payment successfully confirmed by the gateway and available for refund/settlement logic.
Idempotency	The property that retrying the same operation produces one effective result.
KYC	Know Your Customer/Professional identity and document verification process.
Price snapshot	The exact service, tax, fee, discount and policy values retained with a booking.
Service zone	A configured geographic area in which services and professionals are available.
Slot hold	A temporary expiring reservation of capacity before booking confirmation.
Settlement	Transfer of eligible professional earnings for a defined period.
Status history	Append-only record of booking lifecycle transitions.

Appendix B. Suggested Demo Data

Data group	Recommended seed
Cities	Gurugram, Delhi, Noida, Bengaluru.
Categories	Salon for Women, Salon for Men, Spa and Massage, Hair Care, Skin Care, Nail Care.

Data group	Recommended seed
Services	At least 20 services with variants, add-ons, durations and location-specific prices.
Customers	10 seeded customer profiles with addresses and booking histories.
Professionals	12 professionals with varied skills, ratings, zones, availability and KYC status.
Bookings	At least one record in every primary and exception state.
Admin users	Super Admin, Operations Manager, Support Executive, Finance Manager and Catalog Manager.
Payments	Successful, failed, pending, partially refunded and fully refunded examples.
Support	Open, escalated, waiting and resolved ticket examples.

Recommended portfolio demonstration

Use four demo accounts (customer, professional, operations manager and super admin) and a scripted scenario that shows booking, payment verification, assignment, OTP service execution, review, cancellation/refund, and audit history.